

Mathematical Induction II

Principle of Mathematical Induction

● *Proof by mathematical induction:*

To prove that a statement that assert that some property $P(n)$ is true for all positive integers n , we complete two steps

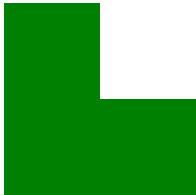
***Basis step:** We verify that $P(1)$ is true.*

***Inductive step:** We show that the conditional statement
 $P(k) \rightarrow P(k + 1)$ is true for all positive integers k*

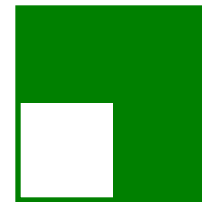
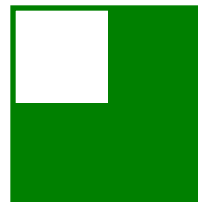
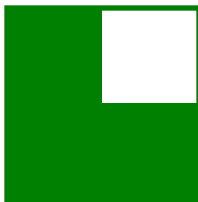
● *To prove the conditional statement, we assume that $P(k)$ is true (it is called **inductive hypothesis**) and show that under this assumption $P(k + 1)$ is also true*

Example (Triomino tiling)

- Let n be a positive integer. Show that every $2^n \times 2^n$ checkerboard with one square removed can be tiled using triominos



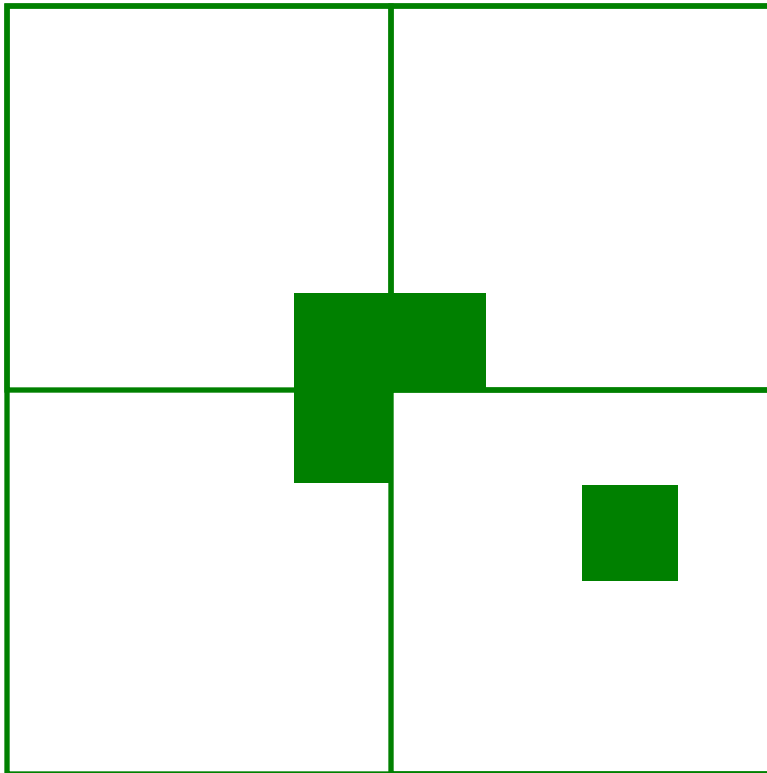
- $P(n)$ denotes the statement above
- Basis step: $P(1)$ is true, as 2×2 checkerboards with one square removed have one of the following shapes



Triomino (cntd)

● *Inductive step: Suppose that $P(k)$ is true that is every $2^k \times 2^k$ checkerboard with one square removed can be tiled with triominos.*

We have to prove $P(k + 1)$, that is, every $2^{k+1} \times 2^{k+1}$ checkerboard without one square can be tiled.



Split the big checkerboard into 4 half-size checkerboards

WLOG assume the bottom right is missing a square. Put one triomino as shown.

We have 4 $2^k \times 2^k$ checkerboards, each without one square. By the induction hypothesis, they can be tiled.

Match game

- *The game starts with two piles of matches*



- *The two players take turns removing any number of matches from a pile of their choice*
- *The player **who takes the last match wins***
- *Theorem:*
if the two piles are the same size initially, then the second player always has a winning strategy

Match game



- Let $P(n)$ denote “player 2 wins if both piles have size n ”

- Base case:** $P(1)$
Player two wins by default



- Inductive step:** Assume $P(k)$
Consider a game with piles of size $k+1$ initially.
Suppose player 1 removes 1 match from the left pile
Then player 2 removes 1 from the right to give two piles of size k



What if player 1 removes more than 1 match?

Principle of Strong Induction

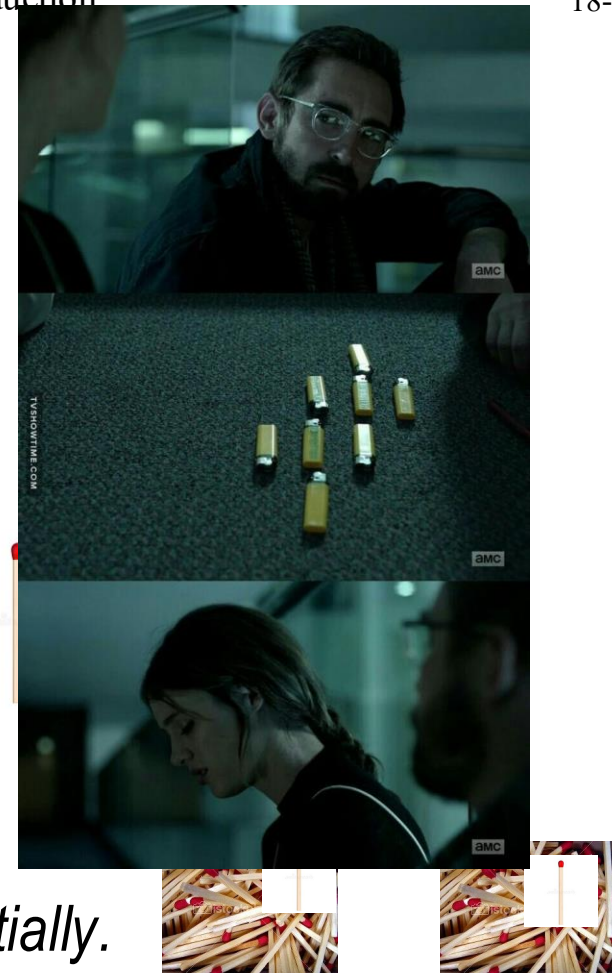
- *In the last proof mathematical induction is not enough. We need to know that the game is winnable for piles of **any** (smaller) size*
- *In these instances we can use the principle of **strong induction**.*
- *Proof by strong induction: To prove that $P(n)$ is true for all positive integers n , we complete two steps:*
 - ***Basis step:** Verify that $P(1)$ is true.*
 - ***Inductive step:** Show that the statement*
$$[P(1) \wedge P(2) \wedge \dots \wedge P(k)] \rightarrow P(k + 1)$$
is true for all positive integers k .

Match game (redux)

● Let $P(n)$ denote "both piles have size n "

● **Base case:** $P(1)$
 Player two wins by default

● **Inductive step:** Assume $P(i)$ for all i up to k
 Consider a game with piles of size $k+1$ initially.
 Suppose player 1 removes x matches from the left
 Then player 2 removes x from the right to give two piles of size $k+1-x$
 Since $x > 0$, $k+1-x$ is at most k , so by the inductive hypothesis player 2 has a winning strategy



Why Induction Works? Well Ordering

- The positive integers are *well-ordered*:

Every non-empty subset of $\mathbb{N}$ contains a least element.

- Are the integers well ordered? Real numbers?

- Theorem: *well ordering implies the principle of induction*

Suppose that mathematical induction is not valid.

Then there is a predicate $P(n)$ such that $P(1)$ is true,

$\forall k (P(k) \rightarrow P(k + 1))$ is true, but there is n such that $P(n)$ is false

Let $T \subseteq \mathbb{N}$ be the set of all n such that $P(n)$ is false.

By the principle of well-ordering T contains a least element k

As $P(1)$ is true, $k \neq 1$.

Since k is the least element of T , $P(k - 1)$ is true

But this implies $P(k)$ is true, hence a contradiction!

Applications of induction: Recursively Defined Functions

- *Induction can also be used to define things.*
- *In this case it's called a **recursive** definition*
- *To define a function $f: \mathbb{N} \rightarrow \mathbb{R}$ recursively we complete two steps:*
 - Basis step:** define $f(1)$*
 - Inductive step:** For all k define $f(k + 1)$ as a function of $f(k)$,
or, more generally, as a function of $f(1), f(2), \dots, f(k)$.*
- *Give a recursive definition of $f(n) = 2^n$*
 - Basis step:** $f(0) = 1$*
 - Inductive step:** $f(k + 1) = 2 \cdot f(k)$.*

Factorial

● *Another useful recursively defined function is factorial*

● $f(n) = n!$

Basis step: $0! = 1$

Inductive step: $(k + 1)! = k! \cdot (k + 1)$

n	$n!$
0	1
1	1
2	2
3	6
4	24

n	$n!$
5	120
6	720
7	5040
8	40320
9	362880

Fibonacci Numbers

- Usually, Fibonacci numbers are thought of as a sequence of natural numbers, but as we know such a sequence can also be viewed as a function from $\mathbb{N}$.
- $F(n)$
- Basis step: $F(1) = F(2) = 1$
- Inductive step: $F(k + 1) = F(k) + F(k - 1)$



n	1	2	3	4	5	6	7	8	9	10	11	12	13
$F(n)$	1	1	2	3	5	8	13	21	34	55	89	144	233

Binet's formula

$$F(n) = \frac{\varphi^n - (1 - \varphi)^n}{\sqrt{5}}$$

where φ is the **golden ratio**

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618033988749$$

Applications of induction: Recursively Defined Sets and Structures

- *Induction can also be used to define **structures***
- *We need to complete the same two steps:*
 - Basis step:** Define the simplest structure possible*
 - Inductive step:** A rule, how to build a bigger structure from smaller ones.*

Well Formed Propositional Statements

- *What is a well formed statement?*

$(p \rightarrow q) \wedge \neg r$ is well formed

$(p \rightarrow q) \neg \wedge r$ is not

- *Recursive definition of well formed formulas*

Basis step: A primitive proposition is a well formed formula

Inductive step: If Φ and Ψ are well formed statements, then

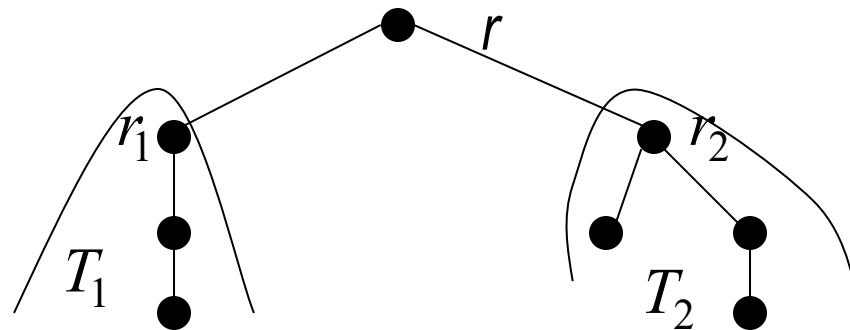
$\neg \Phi$, $(\Phi \wedge \Psi)$, $(\Phi \vee \Psi)$, $(\Phi \rightarrow \Psi)$, $(\Phi \leftrightarrow \Psi)$, $(\Phi \oplus \Psi)$

are well formed statements

- *Computer programs use recursive definitions like these to define most **data structures***

Rooted Trees

- A *binary tree* is a graph formed by the following recursive definition
 - Basis case: A single vertex or nothing is a binary tree ● r
 - Inductive step: Suppose that T_1, T_2 are disjoint binary trees with roots r_1, r_2 , respectively. Then the graph formed by starting with a root r , and adding an edge from r to each of the vertices r_1, r_2 , is also a binary tree.



Structural Induction

- *To prove properties or design algorithms working with recursively defined structures we need **structural induction***
- *To prove a statement using structural induction we complete two steps*
 - Basis step: Prove that the property is true for the simplest structure*
 - Inductive step: Assuming that the property is true for all simpler structures, prove it for a more complex structure*

Structural Induction (cntd)

- Let $h(T)$ be the height of a binary tree, defined recursively as:
 - Basis step: The height of a single vertex r is 0. $h(r) = 0$
 - Inductive step: If a tree T is built from trees T_1, T_2 as shown in the inductive step, then

$$h(T) = 1 + \max(h(T_1), h(T_2))$$

- We prove that the number of vertices in a binary tree, $n(T)$, satisfies the inequality $n(T) \leq 2^{h(T)+1} - 1$

- Basis step: For a single vertex $1 = n(r) \leq 2^{0+1} - 1 = 1$

- Inductive step: Let T be formed from T_1, T_2

We have

$$\begin{aligned}
 n(T) &= 1 + n(T_1) + n(T_2) \\
 &\leq 1 + (2^{h(T_1)+1} - 1) + (2^{h(T_2)+1} - 1) \\
 &\leq 1 + 2(2^{\max(h(T_1), h(T_2))+1} - 1) = 1 + 2^{h(T)+1} - 2 \\
 &= 2^{h(T)+1} - 1
 \end{aligned}$$

Practice

Exercises from the Book:

7th edition: 8, 12 (page 342), 1b, 3a, 7b, 26 (page 357 – 358)

8th edition: 8, 12 (page 363), 1b, 3a, 7b, 28 (page 378 – 379)